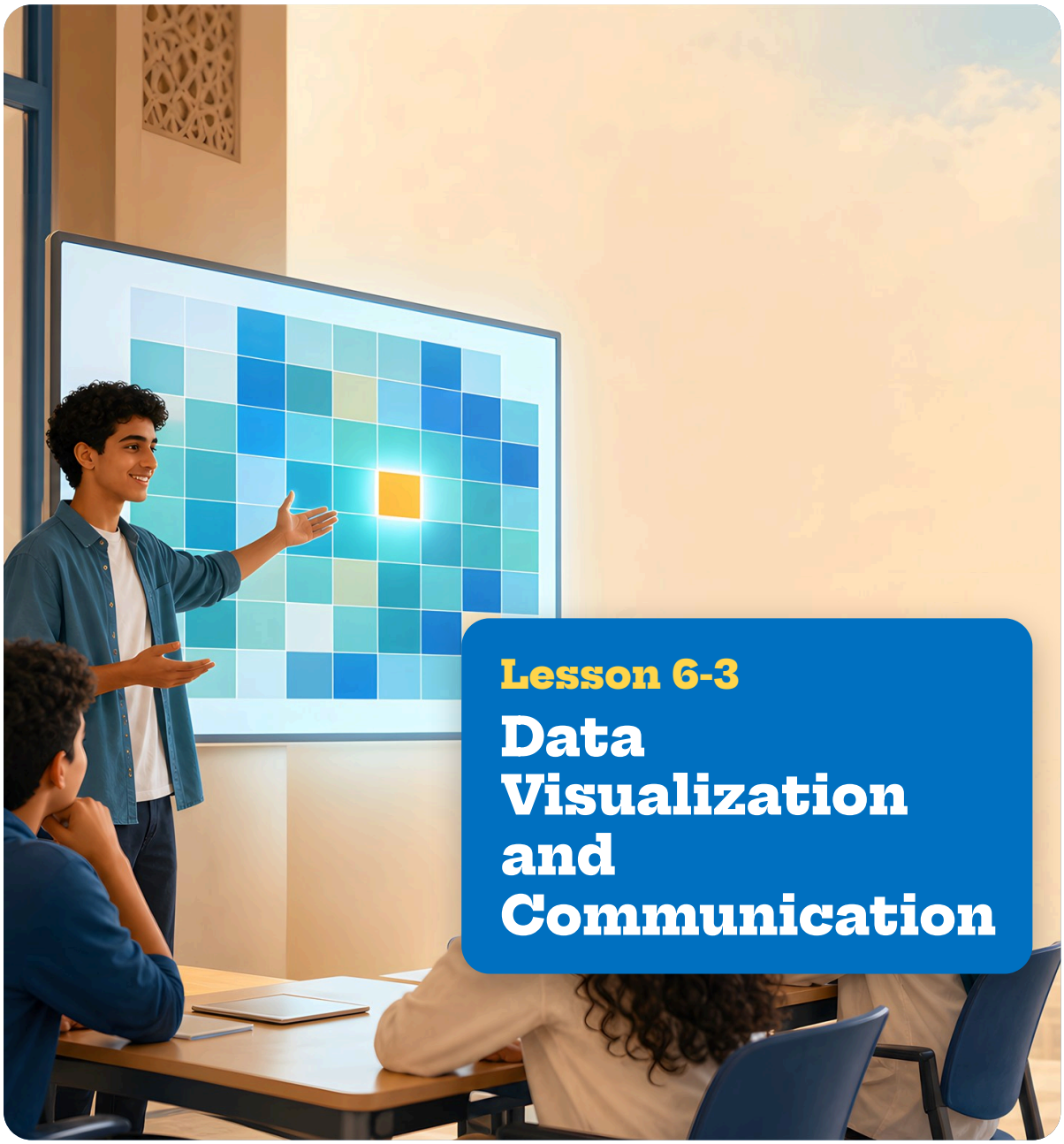


Lesson 6-3

Data Visualization and Communication

Learning Objectives

- Explain the role of data visualization and select suitable graphs, including for multiple variables.
- Apply data storytelling to communicate a clear and honest message from data.

A table of numbers can hide the very thing you most need to see. Turn it into the right picture and the pattern jumps out — the busiest hours, the strongest link, the odd outlier. Visualization does two jobs: it helps you understand your own data, and it helps you explain it to others. When one or two variables are involved, a bar, line, or scatter plot is enough; when three or more variables, or many combinations of information, must be shown at once, heatmaps and mosaic plots are used instead. And a chart alone is not a message: telling a clear data story is what makes an audience actually understand.

? Guiding question: How do we choose a visualization that reveals a pattern honestly, and how do we turn a chart into a message others understand?

Key Fact:
Visualization turns numbers into patterns the eye can grasp — to help the analyst think and to communicate. Common graphs suit one or two variables; heatmaps and mosaic plots handle more combinations. Good communication uses data storytelling: a clear message, only the elements that serve it, and honest labels.

Key Concepts:

- data visualization
- heatmap
- mosaic plot
- overplotting
- data storytelling

Point!

1. The Role of Data Visualization

(1) **(Data visualization)**…representing data as graphs or figures so that features that are hard to read from numbers alone can be grasped visually.

(2) Data visualization broadly has two roles.

Role	Description	Example
Deepen the analyst's own understanding	Visually grasp the distribution, trends, outliers, etc. of data and gain insight	When training records or study time are shown as a line graph, changes and trends that are hard to notice when numbers are merely listed in a table become easier to see
Communicate data to others	Convey the analysis results or claims to others in an easy-to-understand way	When the average test score of each class is shown as a bar graph, even someone seeing the data for the first time can compare differences between classes at a glance

(3) In either role, it is important to choose a graph that fits the content to be conveyed and to use representations that do not invite misunderstanding. The appropriate visualization method differs depending on the number and type of variables and the purpose of the analysis. When information that cannot be handled by familiar graphs needs to be shown, more advanced visualization techniques are used.

KEY TERMS
data visualization — showing data as graphs/figures to grasp features visually.



A clear chart speaks; a cluttered one hides.

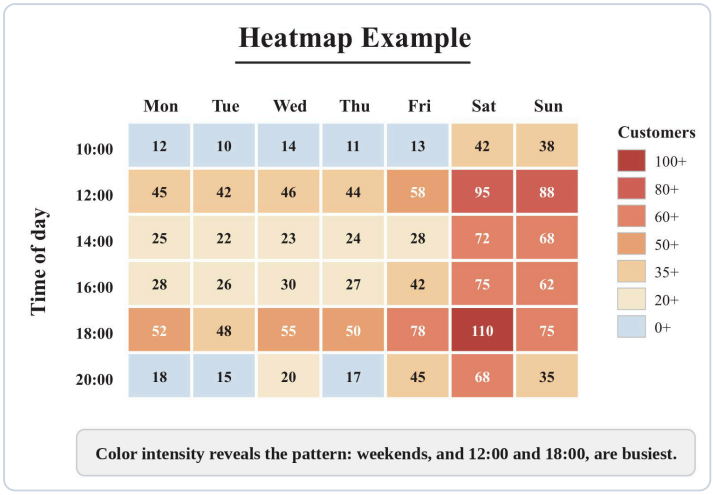
2. Visualization Handling Multiple Variables

Common graphs such as bar graphs, line graphs, and scatter plots are suited to representing the relationship between one or two variables. On the other hand, in analysis, situations often arise in which three or more variables, or many

KEY TERMS
heatmap — a color grid on two axes for a third value.
overplotting — points overlapping so density is hidden.

combinations of information, need to be handled at once. Heatmaps and mosaic plots are advanced visualization methods used for such cases.

(1) **(Heatmap)**…a graph in which items are arranged on two axes, and the size of a **(third value)** — a numerical value — corresponding to each combination is represented by the gradation of color or differences in hue. *e.g.,* Grasp the number of customers visiting at a glance by day of week × time slot; compare trends in average temperature for city × month.



A heatmap: color intensity shows the value for each row × column combination

In a scatter plot, since the relationship between two quantitative variables is represented by points, when the number of points becomes large, they overlap and the density of data becomes hard to see (overplotting). Also, scatter plots are not suited to handling combinations of categorical variables. In such cases, a heatmap has the advantage of allowing the density and patterns of the data to be grasped at a glance.

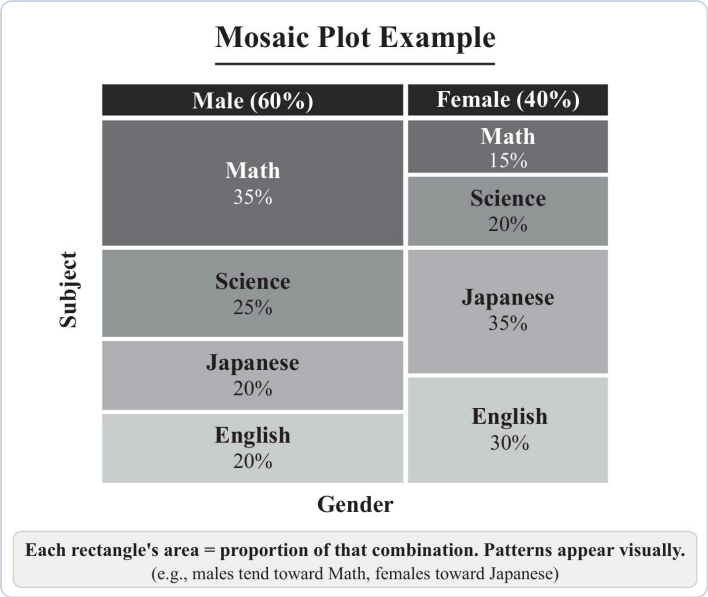
(2) **(Mosaic plot)**…a graph that represents the relationship between two categorical variables by the size of rectangular areas. *e.g.,* Compare the distribution of gender × subject preference; grasp differences in purchase ratios for age group × product category.



One view can encode several variables at once.

PAUSE & THINK

Two categories compared by area, or a third value by color — which graph is which?



A mosaic plot: each rectangle's area shows the proportion of that category combination

The widths of row and column divisions correspond to the proportions of frequencies, and the area of each rectangle represents the proportion of frequency for that combination of categories. Even when the overall picture is hard to grasp from a numerical table (cross-tabulation), a mosaic plot visually shows differences in proportions as areas, making it easier to grasp relationships between categories intuitively.

3. Data Storytelling

(1) **(Data storytelling)**…using data and visualization to convey the message you want to convey, structured like a story.

(2) The following are the three basic elements of storytelling.

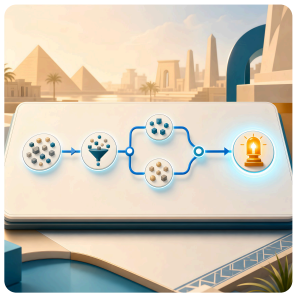
Element	Description	Example
Clarification of the message	Make it possible to express in one sentence what you want to convey through the graph	Make the title of a graph showing class test results contain the claim, such as “The class scored highly overall on the math test”
Selection of elements	Remove elements unrelated to the message and emphasize the parts you want to draw attention to	Make only the elements you want to draw attention to stand out in color, while keeping other elements in faded colors
Provision of supporting information	Properly arrange the title, axis labels, units, and notes	Make the graph title, such as “Average math test scores by class,” express the content at a glance, and clearly indicate units and item names like “points” on the vertical axis and “class” on the horizontal axis

(3) The depth of an audience's understanding can change greatly with how the same analysis result is conveyed. After visualizing data, it is important to put yourself in the audience's position and verify whether the message is conveyed.

KEY TERMS

mosaic plot — rectangle areas for two categorical variables.

data storytelling — structuring a data message like a story.



Storytelling guides the eye to one insight.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Data visualization has both the role of deepening the analyst's own understanding and the role of communicating data to others.

B A heatmap is a graph that represents the size of a third value corresponding to combinations on two axes by the gradation of color or differences in hue.

C A mosaic plot is a graph that represents the relationship between two categorical variables by the size of rectangular areas.

D In data storytelling, it is important to include as many elements related to the graph as possible.

(2) Match each description (a–c) with the most appropriate graph from the choices below (A–C).

a A graph that represents the size of a numerical value corresponding to combinations on two axes by the gradation of color or differences in hue

b A graph that represents the relationship between two categorical variables by the size of rectangular areas

c A graph that represents the relationship between two quantitative variables by the position of points

[Choices] **A** Scatter plot **B** Heatmap **C** Mosaic plot

✓ Solution

(1)

A Correct as a role of data visualization. Therefore, ○.

B Correct definition of a heatmap. Therefore, ○.

C Correct definition of a mosaic plot. Therefore, ○.

D In data storytelling, it is important to remove elements unrelated to the message. Therefore, ×.

(2)

a: Showing values by gradation of color or hue is a heatmap. **B**

b: Showing relationships by area is a mosaic plot. **C**

c: Showing relationships by points is a scatter plot. **A**

PAUSE & THINK

Does adding more elements to a graph make the message clearer, or bury it?

Try

1 Answer the following questions.

1. What is the term for the graph that represents the size of a third value corresponding to combinations on two axes by the gradation of color or differences in hue?
2. What is the term for the graph that represents the relationship between two categorical variables by the size of rectangular areas?
3. In data visualization, of “deepening the analyst's own understanding” and “communicating data to others,” for what purpose is the latter performed?
4. What is the term for using data and visualization to convey the message you want to convey, structured like a story?

2 Answer the following questions.

(1) For each of situations (a, b), choose the most appropriate graph from the choices below (A, B).

a You want to grasp the pattern of customer visits by day of week × time slot at a glance.

b You want to intuitively show differences in proportions for gender × subject preference using areas.

[Choices] **A** Heatmap **B** Mosaic plot

(2) From the following options (A–D), choose the one that most appropriately describes a situation in which a heatmap is used.

A A situation in which one wants to simply compare a representative value for one variable.

B A situation in which the relationship between two quantitative variables can be sufficiently grasped with a scatter plot because the number of points is small.

C A situation in which one wants to grasp the distribution and patterns of numerical data over combinations on two axes at a glance.

D A situation in which one wants to show only one composition ratio of a categorical variable.

EXAM-STYLE QUESTION

Explain why choosing the wrong graph, or hiding a chart's labels, can mislead an audience even when the data is correct. **[6] Refer to:** matching the graph to the data and honest supporting information.

PAUSE & THINK

Two axes with a value by color, or two categories by area? Name each situation's graph.

PAUSE & THINK

Time-series of one value, or proportions of two categories — which suits a mosaic plot?

(3) From the following options (A–D), choose the one that most appropriately describes a situation in which a mosaic plot is used.

A A situation in which one wants to follow the time-series changes of one quantitative variable.

B A situation in which one wants to intuitively grasp the proportions of frequencies for combinations of two categorical variables using areas.

C A situation in which one wants to represent the correlation between two quantitative variables by points.

D A situation in which one wants to show the composition ratio of categories at a particular time using sectors.

(4) From the following options (A–D), choose the one that most appropriately describes data storytelling.

A Even if the message you want to convey is unclear, it will be communicated to the audience as long as the graph is beautiful.

B Clarification of the message, selection of elements, and provision of supporting information are the basic elements.

C It is better to include as many related elements as possible in a graph.

D Titles and axis labels detract from the appearance of a graph and should preferably be omitted.

⚙ Think as an Engineer — Investigate & Decide

Your class collected data on students' home governorate and their favorite sport, and also monthly rainfall for several Egyptian cities across the year. You must present each dataset clearly.

☐ **Use a bar graph for everything**

☐ **Match each dataset to the right graph**

1 · Choose a graph for each. For the governorate × favorite-sport data and for the city × month rainfall data, name the graph you would use. For at least one of them, justify your choice — could the other graph type also work here, and why is yours better?

2 · Guard against misleading. Name one design choice that could make your chosen chart mislead the reader (for example a color scale with no legend, or truncated categories), and say how you would prevent it — a judgment the lesson's examples do not hand you.

3 · Tell the story. For one of the two charts, write the one-sentence message you want the audience to take away, and name the single element you would emphasize to carry it. **In pairs,** compare messages and agree on the clearer one.

Give evidence for your answer.

KEY TERMS

cross-tabulation — a table of counts for category combinations.

misleading graph — one whose design distorts the true pattern.

Stuck? Ask what each dataset is made of: two categories → areas (mosaic); a value across two axes → color grid (heatmap). Then ask what one sentence you want the reader to leave with, and strip away everything that does not serve it.

🌍 In a New Context

A researcher plots study time against test score for every student in a large school as a scatter plot, but so many points overlap into a solid blob that the density cannot be read. Name a visualization that would reveal where most students actually fall, and explain why it works better here than the crowded scatter plot.

❓ Lesson Question — Answered

The lesson asked how to choose a visualization that reveals a pattern honestly and how to turn a chart into a message others understand. Data visualization serves two roles — deepening the analyst's own understanding and communicating to others — and the right method depends on the number and type of variables. Bar, line, and scatter plots suit one or two variables; when a scatter plot's points overlap (overplotting) or categories must be combined, a heatmap shows a value across two axes by color, and a mosaic plot shows two categorical variables by rectangular area. Turning a chart into a message is data storytelling, built from three elements: clarify the message in one sentence, select only the elements that serve it (emphasizing the key part, fading the rest), and provide honest supporting information (title, axis labels, units, notes). Because the same result can land very differently depending on how it is shown, the final step is to stand in the audience's position and check that the message truly arrives.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Data visualization has both the role of deepening the analyst's own understanding and the role of (a) data to others. When information cannot be

handled by common graphs such as bar graphs and line graphs, advanced visualization techniques are used. The graph that represents the size of a third value corresponding to combinations on two axes by the gradation of color or differences in hue is called (b), and the graph that represents the relationship between two categorical variables by the size of rectangular areas is called (c). When communicating data, the concept of (d), in which the message is structured like a story, is important; the basic elements are clarification of the message, selection of elements, and provision of (e).

1. Fill in the blanks (a)–(e).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes a heatmap.

A A graph that represents the relationship between two quantitative variables by the position of points.

B A graph that represents the size of a third value corresponding to combinations on two axes by the gradation of color or differences in hue.

C A graph that represents the composition ratio of categories by the area of sectors.

D A graph that represents the changes in time-series data by lines.

(2) From the following options (A–D), choose the one that is NOT an appropriate basic element of data storytelling.

A Make it possible to express in one sentence what you want to convey through the graph (clarification of the message).

B Remove elements unrelated to the message and emphasize the parts you want to draw attention to (selection of elements).

C Properly arrange the title, axis labels, units, and notes (provision of supporting information).

D Include as many elements unrelated to the message as possible to increase the amount of information.

(3) For each of analysis purposes (a–c), choose the most appropriate graph from the choices below (A–C).

a You want to grasp the monthly average temperatures of multiple cities together at a glance.

b You want to show differences in purchase ratios for age group × product category using areas.

c You want to see the relationship between two quantitative variables in a case where the number of points is small enough that the density can also be sufficiently read.

[Choices] **A** Scatter plot **B** Heatmap **C** Mosaic plot

REFLECT & CHALLENGE

Reflect: Recall a chart that confused you. Was it the wrong graph, too many elements, or missing labels? **Challenge:** You are given a heatmap whose color scale does not start at zero, so small differences look dramatic. Describe the false impression this creates, and the one change that would make it honest without changing the data.

Key takeaway

Remember:
Match the graph to the data (heatmap for a value across two axes; mosaic plot for two categories by area; scatter/bar/line for one or two variables), then tell a data story: one message, only the elements that serve it, honest labels — and check it lands with the audience.